Module 12: Darwin and Evolution - Learning Questions

1. What are the required ingredients for natural selection?
2. Why does natural selection act on existing variation?
3. What does evolutionary fitness mean?
4. Why do populations evolve rather than individuals?
5. How do fossils support evolution?
6. How do homologous and vestigial structures support common ancestry?
7. How can DNA comparisons support relatedness?
8. What is descent with modification?
9. How can environment change which traits are favored?
10. How did Galápagos observations help Darwin reason about adaptation?
11. How can convergent evolution and artificial selection clarify natural selection?
12. Why is adaptation not the same as individual effort?
13. What evidence would support an adaptation claim?
14. How can a trait be helpful in one environment but harmful in another?
15. How should an evolutionary explanation connect variation, heritability, and reproduction?
16. In a fictional insect population, dark and light forms are heritable. Birds remove more light forms on dark rock. What further reproductive evidence would support a natural-selection explanation?

17. Two unrelated species both have streamlined bodies. Why does this alone not establish recent common ancestry, and what additional evidence would you compare?